INSTITUTO
UNIVERSITÁRIO
DE LISBOA

Hierarchical Topic Extraction for Educational Content Representation

Luiza Vidal Copolillo Coelho

Master's in Computer Science and Business Management

Supervisors:
Doctor Ana Rita Peixoto, Associate Professor,
Iscte – Instituto Universitário de Lisboa

2025/26

[This page is intentionally left blank.]

TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

Hierarchical Topic Extraction for Educational Content Representation

Luiza Vidal Copolillo Coelho

Master's in Computer Science and Business Management

Supervisor:
Doctor Ana Rita Peixoto, Associate Professor,
Iscte – Instituto Universitário de Lisboa

2025/26

[This page is intentionally left blank.]

Contents

List of Figures iii

List of Tables v

List of Acronyms vii

Chapter 1. Introduction 1
- 1.1. Contextualization 1
- 1.2. Research Questions and Goals 1
- 1.3. Methodology 1
- 1.4. Document Structure 2

Chapter 2. State-of-the-Art 3
- 2.1. Research Methodology 3
- 2.2. Topic Modeling 8
- 2.3. Knowledge Representation 10
- 2.4. Topic Modeling and Knowledge Representation 11
- 2.5. Challenges in Modeling Educational Content 12

Chapter 3. Methodology 15

References 17

[This page is intentionally left blank.]

List of Figures

Figure 2.1 Pie Chart of Selected Studies Based on Study Type — Count and Percentage Distribution 7

Figure 2.2 Line Chart of Selected Studies Based on Study's Publication Year 7

[This page is intentionally left blank.]

List of Tables

Table 2.1	Stages of an SLR [4]	3
Table 2.2	Conducting the Review Stages	5
Table 2.3	Filters applied to the search	5
Table 2.4	Definition of Keywords and Search String	5
Table 2.5	Final SLR Table — Quantitative Analysis of the Results	6
Table 2.6	Total count of documents — with Snowballing Results	7

[This page is intentionally left blank.]

List of Acronyms

SLR: Systematic Literature Review

RQ: Research Question

CRISP-DM: CRoss Industry Standard Process for Data Mining

TM: Topic Modeling

HTM: Hierarchical Topic Modeling

KR: Knowledge Representation

KG: Knowledge Graph

KW: Keyword

SS: Search String

ML: Machine Learning

LDA: Latent Dirichlet Allocation

PLSA: Probabilistic Latent Semantic Analysis

TF-IDF: Term Frequency-Inverse Document Frequency

BoW: Bag of Words

LSA: Latent Semantic Analysis

IoT: Internet of Things

hLDA: hierarchical Latent Dirichlet Allocation

CRP: Chinese Restaurant Process

GHTM: Guided Hierarchical Topic Model

GCTM: Graph Convolutional Topic Model

QA: Question Answering

KGE: Knowledge Graph Embeddings

CHAPTER 1

Introduction

1.1. Contextualization

In the educational context, written tests remain one of the most prominent forms of student assessment. Despite growing interest in alternative evaluation methods, the process of designing, validating, and reviewing traditional assessments remains labor-intensive and often lacks tools for systematic quality assurance [1]–[3]. This challenge is especially evident in the alignment between test questions and course content, which nowadays is commonly conveyed through lecture slides.

To address these issues, the broader project **TestMATE**—Test Making Assistance a Tool for Education—proposes the development of an intelligent support system for test creation. One of its major components involves automatically extracting and representing the knowledge embedded in lecture slides, this task is naturally complex, given the unstructured nature of slide content and the lack of narrative continuity. Within this scope, this research focuses on investigating how Topic Modeling (TM) techniques, especially Hierarchical Topic Modeling (HTM), can be applied to capture and represent educational content effectively. By exploring the integration of TM with Knowledge Representation (KR), we aim to contribute to the automated understanding of curricular content.

1.2. Research Questions and Goals

The central objective of this thesis is to explore the possible benefits of extracting and structuring educational content using advanced TM techniques. In particular, this research seeks to answer the following research question:

- **RQ1:** Is it possible to extract and represent the knowledge of educational content in the format of slides, using topic modeling?

This research aims to evaluate how hierarchical topic structures can enhance the interpretability and organization of slide-based content and to assess the potential of combining such models with KR techniques, such as ontologies or Knowledge Graphs (KGs), to support next-generation educational applications, such as written evaluation test creation.

1.3. Methodology

To answer the Research Question (RQ) and develop a solid foundation for this study, two methodological approaches are employed. Initially, an Systematic Literature Review (SLR) was conducted to survey existing research on: **Topic Modeling**, **Knowledge Representation**, and their **intersection in educational contexts**. The review

followed established guidelines proposed by Kitchenham [4] and Okoli [5] to ensure scientific methodological reliability.

Additionally, for the practical development and evaluation of the proposed solution, we plan to adopt the CRoss Industry Standard Process for Data Mining (CRISP-DM) methodology, proposed by Wirth and Hipp [6]—further detailed in Chapter 3. This methodology is a potential fit for the development of this research because it allows flexibility and iterative refinement as we explore different TM strategies and KR integration approaches.

1.4. Document Structure

This document is structured as follows. Chapter 1 briefly presents the introduction and scope of the research. Chapter 2 reviews the state-of-the-art on **Topic Modeling**, **Knowledge Representation**, and their **application in the educational domains**. Chapter 3 describes the methodological framework adopted.

CHAPTER 2

State-of-the-Art

2.1. Research Methodology

The methodological approach used to conduct the literature review of this work was a SLR. An SLR is a structured approach to identify, evaluate, and synthesize existing research on a specific topic. Unlike traditional research approaches, SLRs specify precise guidelines and procedures — posteriorly clarified — to reduce bias and guarantee a wide range of relevant studies [7]. In conclusion, this approach improves the scientific reliability and validity of claims conclusions and provides a high-quality foundation for evidence-based practice and further research.

In line with this methodology, the review process was structured according to the phases proposed by Kitchenham [4] and Okoli [5]. The structure of the review revolves around three phases: planning the review, conducting the review, and reporting the review (illustrated in Table 2.1). These phases ensure a structured and repeatable process, aligned with best practices for systematic reviews as defined in software engineering and information systems research.

Table 2.1. Stages of an SLR [4]

Stage	**Activities**
1. Planning the Review	a. Identification of the need for a review b. Development of a review protocol
2. Conducting the Review	a. Identification of research b. Selection of research c. Study Quality Assessment d. Data Extraction and Monitoring e. Data Synthesis
3. Reporting the Review	a. Review Findings and Conclusions

2.1.1. Planning the Review

This section summarizes the first phase of the SLR methodology and clarifies the need for the review, the objective (specification of RQs), and the definition of the review protocol.

2.1.1.1. *Need for the Review*

The main problem with writing assessments today lies in the manual, time-consuming nature of test construction, which often lacks systematic validation and feedback mechanisms [2]. Not only that, but teachers occasionally struggle to ensure alignment between test

content and curricular goals [3]. More specifically, within the scope of this review, a key challenge concerns the representation of educational content, particularly that transmitted through lecture slides, within written assessments designed to evaluate student learning. This issue is further aggravated by practical constraints, including limited preparation time and insufficient institutional support.

Additionally, Sireci [1] points out that the absence of structured methods for collecting content validity evidence further weakens the reliability of learning assessment evaluations. For this reason, this review intends to consolidate the existing research on knowledge representation, and educational topic modeling, as a foundation for developing intelligent tools that can support the automatic identification and structuring of curricular content from lecture slides to enhance test design.

2.1.1.2. *Objective*

The preliminary objective of this research is to provide answers for the following RQ:

RQ1. Is it possible to extract and represent the knowledge of educational content in the format of slides using topic modeling?

2.1.1.3. *Review Protocol*

The review protocol (Table 2.2) began with the definition of keywords based on the research questions defined, which were then combined in different ways to search for publications. The following databases were considered: IEEE Xplore Digital Library[1], ACM Digital Library[2], Scopus[3], and Web of Science[4]. To complement what was presented in 2.1, the SLR methodology involves defining keywords, searching databases, and applying filters to select relevant studies for review. The initial search was performed by searching the search string in the defined databases without applying any filters (i.e., the words in the search string could appear in "Full Text" – or "All Fields", depending on the nomenclature defined in the database. Following that, other filters were applied to further refine the results; Table 2.3 describes all the filters applied.

2.1.2. Conducting the Review

This section describes the second phase of the SLR methodology, detailing the conduction of the review based on the review protocols mentioned above and the analysis of the extracted data.

2.1.2.1. *Identification of the Review*

According to the research questions defined, these were the Keyword (KW) used and combined to form the SS. Besides the KW defined in Table 2.4, synonyms for each word were used to amplify the search and gather more results.

[1]https://ieeexplore.ieee.org/
[2]https://scholar.google.com/
[3]https://www.scopus.com/
[4]https://www.webofscience.com/

Table 2.2. Conducting the Review Stages

Stage	**Activities**
1. Identification of the Research	a. Define KWs based on Title and RQs. b. Define Search String (SS) based on KWs.
2. Study Selection	**Database Search** a. Access the chosen scientific databases. b. Conduct the search with the SS. c. Apply filters. **Snowball Search** a. Select relevant studies from references of the selected studies.
3. Study Quality Assessment	a. Interpret findings carefully.
4. Data extraction and monitoring progress	a. Capture relevant information from studies.
5. Data synthesis	a. Summarize the results of included studies.

Table 2.3. Filters applied to the search

Filter	**Description**
F0	Full Text, All Fields
F1	SK1.1
F2	Only in English
F3	Publication Year $\geq$ 2015
F4	Only Journal Articles or Conference Papers
F5	Subject Area: Comp. Science, Engineering, Business **(applied only in Scopus database)**
F6	Manual Selection
F7	Remove Duplicates

Table 2.4. Definition of Keywords and Search String

Keywords	Topic Model; Hierarchical Topic Model; Knowledge Representation; Knowledge Extraction; Education
SS1	"(("topic model" OR "topic modeling" OR "hierarchical topic model" OR "hierarchical topic modeling") AND("knowledge representation" OR "knowledge extraction" OR "knowledge mining" OR "knowledge graph" OR "knowledge graphs") AND ("education" OR "educational content"))"
SS1.1	"(ABS("topic model" OR "topic modeling" OR "hierarchical topic model" OR "hierarchical topic modeling") AND ABS("knowledge representation" OR "knowledge extraction" OR "knowledge mining" OR "knowledge graph" OR "knowledge graphs") AND ALL("education" OR "educational content"))"

2.1.2.2. *Study Selection*

The first and most important stage of the review process was the database search. A total of 3101 documents were initially retrieved by applying F0 (searching in "Full Text" or "All Fields") across four major digital libraries: IEEE Xplore (434), ACM Digital Library (915), Scopus (1739), and Web of Science (13).

To narrow the scope, the F1 filter applied a predefined keyword string (SS1.1), reducing the dataset to 79 studies. Subsequently, F2 was applied to include only publications written in English, resulting in 75 studies. The third filtering step (F3) further excluded studies published prior to 2015, maintaining a total of 75. Next, F4 limited the selection to Journal Articles or Conference Papers, which again retained 75 studies. Then, F5—**applied exclusively to the Scopus database**—filtered for documents in the subject areas of Computer Science, Engineering, and Business, reducing the dataset to 69 studies.

In the final phase of selection, filters F6 (Manual Selection) and F7 (Duplicate Removal) were applied simultaneously using Zotero tool. In this step, duplicates were merged automatically, and irrelevant documents were manually excluded based on abstract reading. This process reduced the corpus from 64 to 40 studies, and 7 duplicate documents were excluded during manual screening, resulting in a total of 33 primary studies for qualitative synthesis.

This step-by-step approach ensured transparency, reproducibility, and methodological rigor, in line with the SLR guidelines proposed by [4].

In addition to the database search, a Snowballing Search technique is being conducted in parallel throughout the development of this research; this strategy is meant to identify potentially relevant studies that may not have been captured through the initial search queries. As a result, the final number of included articles may be revised in the end. However, studies identified through snowballing will not be considered in the preliminary analysis presented in this chapter, as they were not part of the original systematic filtering process. Table x and x details the quantitative number of studies retrieved from the SLR and the current count of documents from snowballing technique.

Table 2.5. Final SLR Table — Quantitative Analysis of the Results

DB/Filters	**F0**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**
IEEE	434	12	12	12	11	11	9	8
ACM	945	9	9	9	9	9	8	6
Scopus	1739	52	48	48	43	38	20	17
Web of Science	13	6	6	6	6	6	3	2
Total	3101	79	75	75	69	64	40	**33**

2.1.2.3. *Data Extraction and Primary Analysis*

The publications included in the analysis (Figure 2.1) were mainly Journal Articles, comprising 61% of the total, while Conference Papers accounted for the remaining 39%. This

Table 2.6. Total count of documents — with Snowballing Results

Database Search	33
Snowballing Search	9
Total	**42**

distribution indicates a strong level of academic development in the field, as Journal Articles generally require more strict peer-review processes than Conference Papers [8].

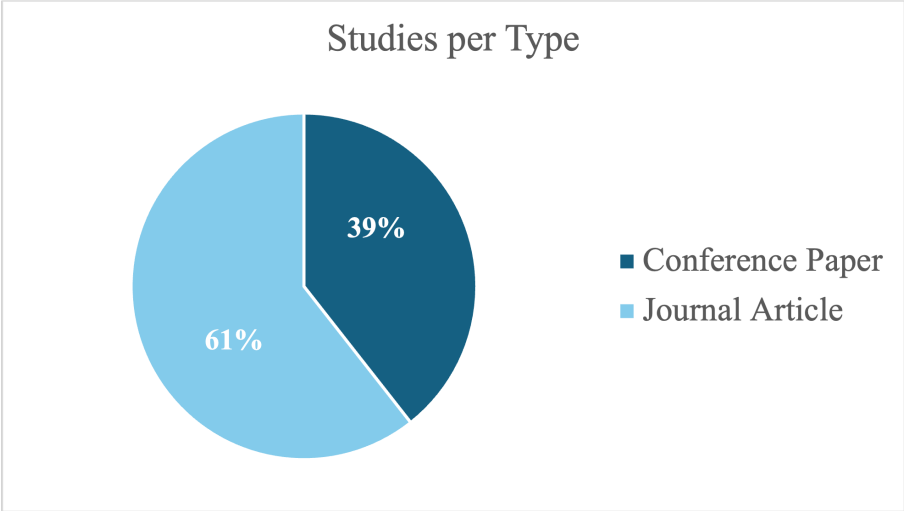

Figure 2.1. Pie Chart of Selected Studies Based on Study Type — Count and Percentage Distribution

Moreover, Figure 2.2 most of the studies were published between 2020 and 2023, with 2023 being the year with the highest number of publications (9 studies). There has been a noticeable increase in research activity since 2019, reflecting a growing interest in this area over the past five years. In contrast, earlier years, especially before 2017, contributed very few studies to the dataset, underlining the recent relevance of the field.

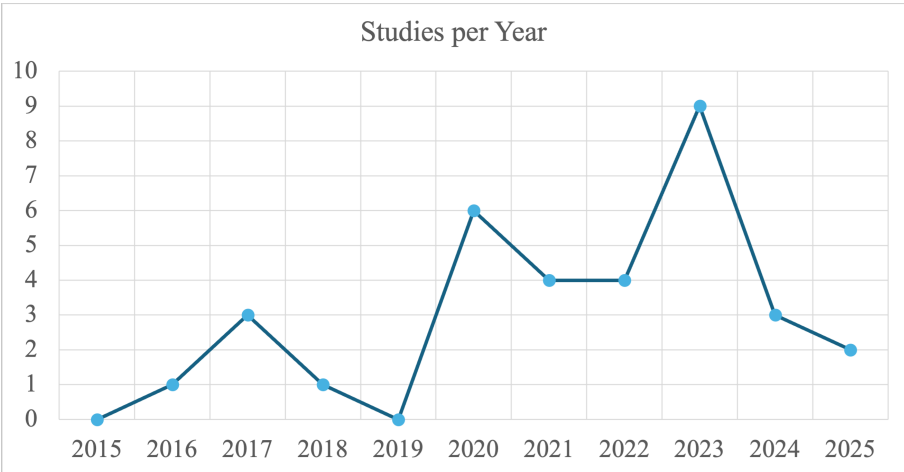

Figure 2.2. Line Chart of Selected Studies Based on Study's Publication Year

2.2. Topic Modeling

TM refers to a combination of unsupervised Machine Learning (ML) techniques used to uncover latent thematic structures within large collections of textual data, while ensuring the correct statistical correlations. It is based on the idea that documents are composed of multiple hidden topics, where each topic is a probability distribution over words, and each document is a mixture of these topics [9], [10]. Unlike traditional vector space models such as Term Frequency-Inverse Document Frequency (TF-IDF) or Bag of Words (BoW), which often result in superficial and semantically shallow representations [11], [12], TM facilitates deeper semantic understanding and more meaningful clustering, classifications, and summarization of textual data.

One of the foundational statistical models in this field is Probabilistic Latent Semantic Analysis (PLSA), introduced by Hofmann [13]. PLSA models the co-occurrence of words and documents through a latent variable framework, in which each word in a document is assumed to be generated from a latent topic variable. While PLSA marked a significant improvement over classical methods like Latent Semantic Analysis (LSA) [14], it lacks a fully generative structure for documents, making it prone to overfitting and incapable of modeling unseen documents without re-training the entire model.

Such limitations were addressed by the development of Latent Dirichlet Allocation (LDA), proposed by Blei, Ng, and Jordan [10]. LDA employs Dirichlet priors [15] to probabilistically model the distribution of topics within documents and words within topics, enabling a generative approach that effectively captures latent structures and generalizes to previously unseen data. More specifically, in LDA, the generative process involves sampling a distribution over topics for each document and, for each word, selecting a topic, and then sampling a word from the corresponding topic distribution. Hence, this structure has made LDA one of the most widely adopted models in various domains such as Media [16], [17], Internet of Things (IoT) [18], Health [19]–[21], Legal [22], [23], Education [24]–[28], and Multidisciplinary [29], [30].

Despite its reliability, LDA and other similar models assume a flat topic structure, which limits their capability when dealing with hierarchical collections. Thus, Griffiths et al. [31] proposed a HTM, named hierarchical Latent Dirichlet Allocation (hLDA), where topics are organized into a tree-like structure that captures specific semantic relationships among them. The hLDA model combines LDA with a nonparametric prior, known as the Chinese Restaurant Process (CRP) [32], which was then extended, and referred to as nested CRP. This prior enables the discovery of topic hierarchies without predefining the number of topics nor the depth of the tree, and in hLDA, each document is modeled as a path through an infinite tree, sampling words from topic distributions associated with nodes along that path. Moreover, a Gibbs sampling algorithm is implemented for inference, allowing the model to simultaneously explore both the topic content and hierarchical structure. On the other hand, these models often constrained each document to

a single path, restricting their capacity to model documents that span multiple thematic areas.

On top of that, Shin and Moon [33] identify another limitation of hLDA hLDA [31]: its inability to integrate prior-knowledge or domain-specific knowledge during the topic inference process, often leading to abstract topic structures that lack semantic coherence. In contrast, Shin and Moon have sought to address this limitation by introducing Guided Hierarchical Topic Model (GHTM), which offers a more robust approach by explicitly merging HTM with structured knowledge representation. GHTM is the first model to apply the Dirichlet Forest prior, an advancement of the traditional Dirichlet prior [15] that incorporates semantic constraints through a forest-like structure of topic paths. This approach allows documents to be associated with multiple topic hierarchies and incorporates prior knowledge through *must-link* and *cannot-link* clustering relationships, guiding the model in grouping semantically related words while effectively distinguishing unrelated ones.

By incorporating these constraints, GHTM represents sort of a hybrid approach of TM and KR (explained in more detail in Section 2.3). This model aligns with broader trends in semantic modeling, where structured knowledge—often represented **through Ontologies or KGs**—is employed to enhance interpretability and relevance of machine-generated topics. The guided facilitates the development of topic hierarchies that not only reflect statistical patterns in the data but also adhere to domain ontologies or expert knowledge, resulting in topics that are both semantically meaningful and pedagogically relevant. Overall, the empirical evaluations suggest that GHTM outperforms traditional HTMs in key metrics such as F-measure, perplexity, and KL-divergence-based separability.

The progression towards more expressive TMs has also led to hybrid frameworks that incorporate external semantic knowledge. For instance, Fatimi et al. [12] proposed a framework for semantic clustering where unstructured text is transformed into RDF triples and enhanced using background knowledge to better capture meaning, supporting better topic interpretation and domain-specific reasoning. Likewise, Linh et al. [34] introduced a Graph Convolutional Topic Model (GCTM) that enhances topic inference for short and noisy text streams-an environment where traditional models often underperform. GCTM incorporates external word knowledge graphs, such as WordNet or embeddings like Word2Vec, and applies graph convolutional layers to strengthen co-occurrence patterns by uncovering deeper semantic connections. This results in more accurate and coherent topic structures, even under concept drift or evolving data.

Moreover, recent advances in topic modeling have embraced neural and transformer-based methods to improve topic coherence and adaptability. As an example, Ennajari et al. [35] proposed a spherical topic model that integrates knowledge graphs and word embeddings, enhancing semantic structure in the latent space. Another work [36] applied few-shot relation learning using transformer architectures to extract meaningful relations from limited data, improving knowledge-aware topic extraction. Additionally, Arslan and

Cruz [37] used BERTopic with controlled vocabularies to model business trajectories, enabling semantically enriched, domain-independent topic representations. Hence, these approaches reflect a trend towards hybrid models that combine structured knowledge with contextual embeddings.

In summary, topic modeling has undergone a considerable improvement—from flat probabilistic models like PLSA and LDA, to hierarchical frameworks such as hLDA and GHTM, and more recently, to hybrid and neural architectures that incorporate semantic knowledge and deep contextual embeddings. These advancements not only overcome previous limitations related to model inflexibility and topic interpretability, but also improve the potential of TMs strategies to function effectively across diverse and complex domains. As a result, modern TMs techniques are currently capable of capturing rich semantic structures, adapting to noisy or evolving corpora, and integrating domain-specific knowledge, thus reinforcing their relevance and applicability in information retrieval, decision-support systems, and knowledge representation.

2.3. Knowledge Representation

According to Nguyen et al. [24], KR refers to the structured modeling of information using logical components that make the semantics of a specific domain or topic machine-interpretable. This is increasingly important for improving the effectiveness of document retrieval systems and supporting different semantic applications. By encoding the meaning, content, and application code of textual data separately, KR allows machines to comprehend textual data, exchange understandings, and make inferences.

Among the most common knowledge structures, Ontologies, and KGs are frequently used. Ontologies provide the formal schema and logical underpinnings, while KGs instantiate these schemas with real-world entities and relationships. Both of them offer a robust practice for organizing, structuring, and analyzing large amounts of data that have complex relationships [19], [38].

2.3.1. Ontologies

An ontology is defined as a shared vocabulary and common information within a specific knowledge domain, encompassing definitions of fundamental concepts that can be interpreted by computers for understanding within that specific field. Ontologies serve as the blueprint for constructing KGs, while KGs capture the actual data, ontologies define the rules and vocabulary that structure this data [12], [24], [39].

Languages such as the Web Ontology Language (OWL) [40] are commonly used to define ontologies, offering a formal semantics grounded in first-order logic and model theory, which allows automated reasoning and inference. For example, in the educational domain, entities like *Course*, *Student*, and *Assessment* may be defined, along with relationships such as *enrolledIn* or *hasGrade*. These formal representations enable semantic interoperability, allowing heterogeneous systems to integrate, share, and reason over domain knowledge effectively.

2.3.2. Knowledge Graphs

A Knowledge Graph (KG) is a network representation of real-world entities (such as objects, events, and people) and the relationships encoded between them [41]. KGs are typically composed of three main components:

- **Nodes:** Represents entities or concepts (e.g, people, events, objects)
- **Edges:** Represents semantic relationships between entities
- **Labels:** Describes Nodes and Edges, enriches the graph with semantic context and enables interpretability by both humans and machines

Additionally, they are often structured as directed heterogeneous graphs where nodes and edges indicate entities and semantic linkages. Therefore, a KG can be described as a collection of triples (h, r, t), where 'h' is the head entity, 't' is the tail entity, and 'r' is the relationship linking them [35]. They organize data from multiple sources and capture information about entities of interest, forging connections between them.

In summary, Ontologies and KGs together form the backbone of modern knowledge representation strategies that enable intelligent systems to effectively utilize and process knowledge information [21]. Ontologies provide the formal, machine-interpretable definitions and semantic structure required to model domain knowledge with precision and consistency. Building upon this foundation, KGs enable the practical organization of real-world data through richly interconnected entities and relationships [12]. Thus, when combined, they enable reasoning, data integration, and semantic interoperability across various sources, enabling intelligent systems to comprehend, relate, and act upon complex information in a meaningful and scalable manner. This can be used to enhance various tasks such as question answering [27], recommendation systems [42], and information retrieval[25].

2.4. Topic Modeling and Knowledge Representation

Following the described background of TM and KR, it is possible to assume the these two fields are profoundly interconnected that, when combined, offer a mroe sophisticated and interpretable understanding and organization of information, moving beyond simple text analysis, to capturing, for instance, rich semantic meaning and relationships [12].

Moreover, this integration enhances the interpretability and semantic coherence of topics discovered by topic models, which can otherwise be unclear or difficult to interpret due to their lack of semantic context or prior knowledge [24]. By incorporating external knowledge from KGs or ontologies, TM can infer more accurate, coherent, and semantically meaningful topics [35], [43]. For instance, Knowledge Graph Embeddings (KGE), which map entities and relationships into low-dimensional vectors while preserving the graph's structure, can guide the inference process of semantically coherent topics. Moreover, hybrid models like BERTopic [37] leverage contextualized language representations to consider underlying semantic context and create denser clusters of interpretable topics.

In conclusion, these works showcase that topic modeling has the potential to be helpful in the construction and enrichment of KGs and ontologies from unstructured text by identifying relevant thematic concepts, key terms, and summarizing content to build structured representations. A few examples include LDA for classifying medical articles into topics for lung cancer knowledge classification [19], or for building KGs to analyze narrative structures in literary works [44].

2.5. Challenges in Modeling Educational Content

The modeling of educational content—particularly for purposes such as knowledge representation, recommendation, or test construction—presents a complex array of challenges that span semantic ambiguity, structural inconsistency, and the lack of ground-truth evaluation [1]–[3].

A recurrent challenge in the context of education is the absence of a structured framework oriented to specific domains. Educational resources, including lecture slides and exam questions, frequently exhibit inconsistent formatting and a lack of semantic unity. This inconsistency poses significant challenges for the implementation of conventional topic modeling techniques. In response to these issues, Xue et al. [26] introduced a dynamic educational knowledge graph model that utilizes information entropy to tackle the complexities of knowledge evolution and concept drift within learning environments. Although this model enhanced the adaptability of content representation, it still faced limitations related to the initial coverage of the knowledge base and its reliance on high-quality labeled data. Thus, this indicates a need for further investigation into unsupervised methodologies to overcome these constraints.

Another challenge concerns content sparsity and incoherence in short texts, such as *quiz items* or *slide bullets*. Shin and Moon addressed this by developing Guided Hierarchical Topic Model (GHTM)—previously described in Section 2.2, which integrates domain knowledge through Dirichlet Forest priors to structure topics hierarchically. Their approach demonstrated improved clustering and interpretability; however, it also relied heavily on the quality and specificity of the seed words, making its adaptability to new or interdisciplinary subjects limited. In conclusion, they acknowledged that future research should refine seed word selection strategies and develop automated methods for domain knowledge extraction

Additionally, semantic gap and heterogeneity across learning objects presents a barrier to uniform representation. Yi et al. [28] incorporated sentiment analysis and KGs to enhance e-learning evaluation, revealing how emotionally annotated feedback could contextualize student engagement. Yet, their system struggled to integrate heterogeneous learning resources such as slides, forums, and quizzes due to the lack of a unified semantic schema.

In more specialized domains like science education, Zhou et al. [27] developed a Question Answering (QA) system for chemistry education that mapped questions to structured knowledge graphs. This approach improved precision in response generation, but

faced challenges in scalability and adaptability across diverse subfields within chemistry, highlighting the need for more modular and cross-domain compatible models.

[This page is intentionally left blank.]

CHAPTER 3

Methodology

As introduced in Chapter 1, we plan to follow a CRISP-DM methodology to guide or process. Proposed by Wirth and Hipp [6], the CRISP-DM is a framework that structures data mining projects into phases: (1) business understanding, (2) data understanding, (3) data preparation, (4) modeling, (5) evaluation, and (6) deployment. It emphasizes iterative development and feedback loops between phases. By applying CRISP-DM to our solution, the objective is to document each step thoroughly in order to inform future iterations and incrementally build tasks that address our specific context, such as dealing with multimodal content and integrating TM outputs with KR structures.

We intend to undertake fundamental preprocessing steps to transform lecture slides into machine-readable text, identify potential data quality issues, and prepare initial datasets. Following that, we plan to explore and test HTM techniques preliminary, without committing to a definitive modeling strategy at this stage. Additionally, we aim to qualitatively assess the meaningfulness of the extracted topics and investigate whether the integration of knowledge representation methods, such as simple graphs or ontologies, could enhance semantic coherence.

Finally, as this proposal is in its early stages, we acknowledge that the methodology may undergo modifications. Further iterations of data understanding, modeling, and evaluation will be necessary, in keeping with the iterative and flexible nature of CRISP-DM, to refine our approach as new insights are gained.

[This page is intentionally left blank.]

References

[1] S. G. Sireci, "Gathering and analyzing content validity data," *Educational Assessment*, vol. 5, no. 4, pp. 299–321, 1998.

[2] R. E. Bennett, "Formative assessment: A critical review," *Assessment in education: principles, policy & practice*, vol. 18, no. 1, pp. 5–25, 2011.

[3] S. McCallum and M. M. Milner, "The effectiveness of formative assessment: Student views and staff reflections," *Assessment & Evaluation in Higher Education*, vol. 46, no. 1, pp. 1–16, 2021.

[4] B. Kitchenham, "Procedures for performing systematic reviews," *Keele, UK, Keele University*, vol. 33, no. 2004, pp. 1–26, 2004.

[5] C. Okoli, "A guide to conducting a standalone systematic literature review," *Communications of the Association for Information Systems*, vol. 37, 2015.

[6] R. Wirth and J. Hipp, "Crisp-dm: Towards a standard process model for data mining," in *Proceedings of the 4th international conference on the practical applications of knowledge discovery and data mining*, Manchester, vol. 1, 2000, pp. 29–39.

[7] C. Okoli and K. Schabram, "A guide to conducting a systematic literature review of information systems research," 2015.

[8] M. Montesi and J. M. Owen, "From conference to journal publication: How conference papers in software engineering are extended for publication in journals," *Journal of the American Society for Information Science and Technology*, vol. 59, no. 5, pp. 816–829, 2008.

[9] D. M. Blei, "Probabilistic topic models," *Communications of the ACM*, vol. 55, no. 4, pp. 77–84, 2012.

[10] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *Journal of machine Learning research*, vol. 3, no. Jan, pp. 993–1022, 2003.

[11] T. Wei, Y. Lu, H. Chang, Q. Zhou, and X. Bao, "A semantic approach for text clustering using wordnet and lexical chains," *Expert Systems with applications*, vol. 42, no. 4, pp. 2264–2275, 2015.

[12] S. Fatimi, C. E. Saili, and L. Alaoui, "A framework for semantic text clustering," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 6, 2020.

[13] T. Hofmann, "Probabilistic latent semantic analysis," in *Proceedings of the Fifteenth conference on Uncertainty in artificial intelligence*, Morgan Kaufmann Publishers Inc., Jul. 1999, pp. 289–296.

[14] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," *Journal of the American society for information science*, vol. 41, no. 6, pp. 391–407, 1990.

[15] J. Sethuraman, "A constructive definition of dirichlet priors," *Statistica sinica*, pp. 639–650, 1994.

[16] Y. Xia, X. Wang, L. Gu, Q. Gao, J. Jiao, and C. Wang, "A collective entity linking algorithm with parallel computing on large-scale knowledge base," *The Journal of Supercomputing*, vol. 76, no. 2, pp. 948–963, 2020.

[17] X. Li, Z. Li, and Y. Tian, "Sentimental knowledge graph analysis of the covid-19 pandemic based on the official account of chinese universities," *Electronics*, vol. 10, no. 23, p. 2921, 2021.

[18] Y. Liu, F. Du, J. Sun, Y. Jiang, J. He, T. Zhu, and C. Sun, "A crowdsourcing-based topic model for service matchmaking in internet of things," *Future Generation Computer Systems*, vol. 87, pp. 186–197, 2018.

[19] C.-H. Cheng and Z.-T. Ji, "A weighted-link graph neural network for lung cancer knowledge classification," *Applied Intelligence*, vol. 53, no. 14, pp. 17 610–17 628, 2023.

[20] A. Mechergui, W. B. A. Karaa, and S. Zghal, "Duplication-driven distributional topic modeling: A catalyst for strengthened classification and semantic graphs.," in *TACC*, 2023, pp. 101–107.

[21] M. Ali, S. Lee, and B. H. Kang, "Udekam: A methodology for acquiring declarative structured knowledge from unstructured knowledge resources," in *2016 International Conference on Machine Learning and Cybernetics (ICMLC)*, IEEE, vol. 1, 2016, pp. 177–182.

[22] K. P. Joshi and S. Saha, "A semantically rich framework for knowledge representation of code of federal regulations," *Digital Government: Research and Practice*, vol. 1, no. 3, pp. 1–17, 2020.

[23] A.-M. Niculae, S.-V. Oprea, A.-G. Văduva, A. Bâra, and A.-I. Andreescu, "Assessing the role of machine learning in climate research publications," *Sustainability*, vol. 16, no. 24, p. 11 086, 2024.

[24] D. T. Nguyen, N. V. Do, and T. H. Tran, "A model of topic for document retrieval systems in the field of artificial intelligence for information technology students," in *Proceedings of the 2024 9th International Conference on Intelligent Information Technology*, 2024, pp. 376–385.

[25] J. Sanguino, R. Manrique, and O. Mariño, "A semantic enhanced course recommender system via knowledge graphs for limited user information scenarios," *SN Computer Science*, vol. 5, no. 1, p. 120, 2023.

[26] Z. Xue, H. Liu, Z. Zhang, and S. Yang, "Dynamic educational knowledge graph model via information entropy for knowledge building," in *2021 IEEE International*

Conference on Engineering, Technology & Education (TALE), IEEE, 2021, pp. 01–07.

[27] X. Zhou, D. Nurkowski, S. Mosbach, J. Akroyd, and M. Kraft, "Question answering system for chemistry," *Journal of Chemical Information and Modeling*, vol. 61, no. 8, pp. 3868–3880, 2021.

[28] W. Yi, X. Huang, S. Kuzmin, I. Gerasimov, and Y. Luo, "Seekg: Sentiment analysis for e-learning evaluation incorporating knowledge graphs," *Education and Information Technologies*, pp. 1–30, 2025.

[29] G. Stănescu and S.-V. Oprea, "Recent trends and insights in semantic web and ontology-driven knowledge representation across disciplines using topic modeling," *Electronics*, vol. 14, no. 7, p. 1313, 2025.

[30] Y. Hong, S. Feng, and Y. Xiao, "Entitylda: A topic model for entity retrieval on knowledge graph," in *2020 IEEE International Conference on Knowledge Graph (ICKG)*, IEEE, 2020, pp. 388–395.

[31] T. Griffiths, M. Jordan, J. Tenenbaum, and D. Blei, "Hierarchical topic models and the nested chinese restaurant process," *Advances in neural information processing systems*, vol. 16, 2003.

[32] D. J. Aldous, I. A. Ibragimov, J. Jacod, and D. J. Aldous, *Exchangeability and related topics*. Springer, 1985.

[33] S.-J. Shin and I.-C. Moon, "Guided htm: Hierarchical topic model with dirichlet forest priors," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 2, pp. 330–343, 2016.

[34] N. Van Linh, T. X. Bach, and K. Than, "A graph convolutional topic model for short and noisy text streams," *Neurocomputing*, vol. 468, pp. 345–359, 2022.

[35] H. Ennajari, N. Bouguila, and J. Bentahar, "Combining knowledge graph and word embeddings for spherical topic modeling," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 7, pp. 3609–3623, 2021.

[36] H. Li and B. Liu, "Knowledge extraction: A few-shot relation learning approach," in *2022 International Conference on Machine Learning and Knowledge Engineering (MLKE)*, IEEE, 2022, pp. 261–265.

[37] M. Arslan and C. Cruz, "Modeling semantic business trajectories of territories for multidisciplinary studies through controlled vocabularies," in *2023 IEEE 39th International Conference on Data Engineering Workshops (ICDEW)*, IEEE, 2023, pp. 170–177.

[38] M. Bozhidarova, J. Chang, A. Ale-Rasool, Y. Liu, C. Ma, A. L. Bertozzi, P. J. Brantingham, J. Lin, and S. Krishnagopal, "Hate speech and hate crimes: A data-driven study of evolving discourse around marginalized groups," in *2023 IEEE International Conference on Big Data (BigData)*, IEEE, 2023, pp. 3107–3116.

[39] S. Staab and R. Studer, *Handbook on ontologies*. Springer Science & Business Media, 2013.

[40] G. Antoniou and F. v. Harmelen, "Web ontology language: Owl," *Handbook on ontologies*, pp. 91–110, 2009.

[41] IBM Corporation, *What is a Knowledge Graph?* https://www.ibm.com/think/topics/knowledge-graph, Accessed: 2025-06-25, 2025.

[42] Q. Li, Z. Zhang, F. Zhuang, Y. Xu, and C. Li, "Topic-aware intention network for explainable recommendation with knowledge enhancement," *ACM Transactions on Information Systems*, vol. 41, no. 4, pp. 1–23, 2023.

[43] L. Yao, Y. Zhang, B. Wei, Z. Jin, R. Zhang, Y. Zhang, and Q. Chen, "Incorporating knowledge graph embeddings into topic modeling," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 31, 2017.

[44] X. Li, "Transformation and Innovation of Chinese Modern and Contemporary Literature in the Context of the Digital Age," *Applied Mathematics and Nonlinear Sciences*, vol. 9, no. 1, 2024. DOI: 10.2478/amns-2024-0668. [Online]. Available: https://doi.org/10.2478/amns-2024-0668.